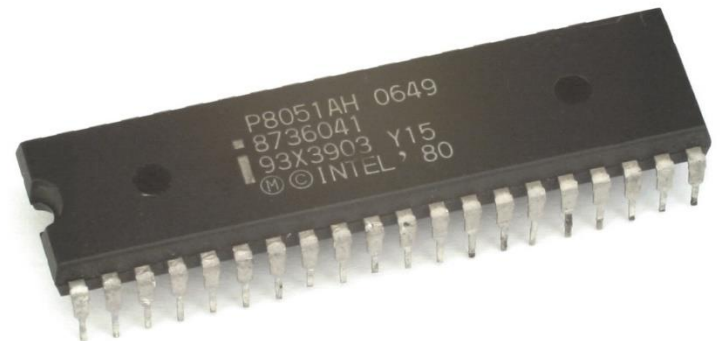


Microprocessor Systems

Appendix – SDCC & Keil

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Difference between SDCC & Keil

- Include header file
 - SDCC needs to include “8051.h”
 - KEIL needs to include <REGX51.h>
- “__” usage
 - SDCC will use “__” sometimes , e.g. __interrupt (?), __bit,...
 - KEIL will not use “__”
- Project and File
 - SDCC : Use makefile & sdcc compiler to make project
 - KEIL : After first compiling of main file, it will add all include file to project

Difference between SDCC & Keil

```
#include "8051.h"
```

SDCC寫法

```
void T0_isr(void) __interrupt (1)
{
    // PC jumps to here every 1ms
    TH0 = (65536-1000) / 256;
    TL0 = (65536-1000) % 256;

    cnt ++;
    if(cnt >= 1000)
    {
        cnt = 0;
        P1 = ~P1;
    }
}
```

SDCC寫法

```
#include <REGX51.h>
```

KEIL寫法

```
void T0_isr(void) interrupt 1
{
    // PC jumps to here every 1ms
    TH0 = (65536-1000) / 256;
    TL0 = (65536-1000) % 256;

    cnt ++;
    if(cnt >= 1000)
    {
        cnt = 0;
        P1 = ~P1;
    }
}
```

KEIL寫法

Difference between SDCC & Keil

```
void I2C_SendACK(__bit ack)
{
    SDA = ack;
    delay_i2c();
    SCL = 1;
    delay_i2c();
    SCL = 0;
    delay_i2c();
}

//*****

//*****

__sbit I2C_RecvACK()
{
    delay_i2c();
    SDA = 1;
    delay_i2c();
    SCL = 1;
    delay_i2c();
    CY = SDA;
    delay_i2c();
    SCL = 0;
    delay_i2c();
    return CY;
}
```

MIDE寫法

```
// Send ACK
void I2C_SendACK(bit ack)
{
    SDA = ack;
    SCL = 1;
    Delay();
    SCL = 0;
    Delay();
}

// Recieve ACK
bit I2C_RecvACK()
{
    SCL = 1;
    Delay();
    CY = SDA;
    SCL = 0;
    Delay();
    return CY;
}
```

KEIL寫法

Difference between SDCC & Keil

```
__code unsigned char const OledFontTable [] = {
    0x00, 0x00, 0x00, 0x00, 0x00, // space
    0x00, 0x00, 0x2f, 0x00, 0x00, // !
    0x00, 0x07, 0x00, 0x07, 0x00, // "
    0x14, 0x7f, 0x14, 0x7f, 0x14, // #
    0x24, 0x2a, 0x7f, 0x2a, 0x12, // $
    0x23, 0x13, 0x08, 0x64, 0x62, // %
    0x36, 0x49, 0x55, 0x22, 0x50, // &
    0x00, 0x05, 0x03, 0x00, 0x00, // '
    0x00, 0x1c, 0x22, 0x41, 0x00, // (
    0x00, 0x41, 0x22, 0x1c, 0x00, // )
    0x14, 0x08, 0x3E, 0x08, 0x14, // *
    0x08, 0x08, 0x3E, 0x08, 0x08, // +
    0x00, 0x00, 0xA0, 0x60, 0x00, // ,
    0x08, 0x08, 0x08, 0x08, 0x08, // -
```

SDCC寫法

```
code const OledFontTable[] =
{
    0x00, 0x00, 0x00, 0x00, 0x00, // space
    0x08, 0x08, 0x08, 0x08, 0x08, // -

    0x3E, 0x51, 0x49, 0x45, 0x3E, // 0
    0x00, 0x42, 0x7F, 0x40, 0x00, // 1
    0x42, 0x61, 0x51, 0x49, 0x46, // 2
    0x21, 0x41, 0x45, 0x4B, 0x31, // 3
    0x18, 0x14, 0x12, 0x7F, 0x10, // 4
    0x27, 0x45, 0x45, 0x45, 0x39, // 5
    0x3C, 0x4A, 0x49, 0x49, 0x30, // 6
    0x01, 0x71, 0x09, 0x05, 0x03, // 7
    0x36, 0x49, 0x49, 0x49, 0x36, // 8
    0x06, 0x49, 0x49, 0x29, 0x1E, // 9

    0x7C, 0x12, 0x11, 0x12, 0x7C, // A
    0x3E, 0x41, 0x49, 0x49, 0x7A, // G
    0x63, 0x14, 0x08, 0x14, 0x63, // X
    0x07, 0x08, 0x70, 0x08, 0x07, // Y
    0x61, 0x51, 0x49, 0x45, 0x43, // Z
};
```

KEIL寫法